

Lava Database Schemas

Lava has **two independent databases**:

1. **Client (Android) — Room/SQLite**, `lava.database.AppDatabase`, schema **version 11**. The authoritative dump is the Room schema JSON at [core/database/schemas/lava.database.AppDatabase/11.json](#) (identity hash 72458bec62ab2a038aeb36fbfe05129d). Every column / type / key below is taken verbatim from that file.
2. **Server (lava-api-go) — Postgres**, schema `lava_api`, managed by `golang-migrate` SQL files [lava-api-go/migrations/0001..0009](#). Every column / type / key below is taken verbatim from the `*.up.sql` files.

These two schemas are **not** mirrors of each other. They overlap on a small number of conceptual entities (provider config/credentials, provider selections) which evolved on both sides; see §3 (Parity) for the overlap.

1. Client schema (Room / SQLite, version 11)

Room affinity → SQLite type: TEXT, INTEGER, BLOB. `notNull: true` = NOT NULL. Nullable columns omit the flag.

1.1 ER diagram (client)

The only declared relationship is `ForumCategoryEntity`'s self-referential `parentId` → `id` foreign key (ON DELETE CASCADE). All other tables are standalone (Room stores related data as serialized TEXT, e.g. `Bookmark.topics`, rather than via FKs).

erDiagram

```
ForumCategoryEntity ||--o{ ForumCategoryEntity : "parentId (CASCADE)"
```

```
Bookmark {
    TEXT id PK
    INTEGER timestamp
    TEXT category
    TEXT topics
}
```

```

        TEXT newTopics
    }
    cloned_provider {
        TEXT syntheticId PK
        TEXT sourceTrackerId
        TEXT displayName
        TEXT primaryUrl
        INTEGER deletedAt "nullable"
    }
    credentials_entry {
        TEXT id PK
        TEXT displayName
        TEXT type
        BLOB ciphertext
        INTEGER createdAt
        INTEGER updatedAt
        INTEGER deletedAt "nullable"
    }
    Endpoint {
        TEXT id PK
        TEXT type
        TEXT host
    }
    FavoriteSearch {
        INTEGER id PK
    }
    FavoriteTopic {
        TEXT id PK
        INTEGER timestamp
        TEXT title
        TEXT author "nullable"
        TEXT category "nullable"
        TEXT tags "nullable"
        TEXT status "nullable"
        INTEGER date "nullable"
        TEXT size "nullable"
        INTEGER seeds "nullable"
        INTEGER leeches "nullable"
        TEXT magnetLink "nullable"
        INTEGER hasUpdate
    }
    ForumCategoryEntity {
        TEXT id PK
        TEXT name
        TEXT parentId FK "nullable"
        INTEGER orderIndex
    }
    ForumMetadata {
        INTEGER id PK
        INTEGER lastUpdatedTimestamp
    }
    forum_provider_selections {

```

```

        INTEGER id PK "autoincrement"
        TEXT category_id
        TEXT provider_id
        INTEGER created_at
    }
    tracker_mirror_health {
        TEXT tracker_id PK
        TEXT mirror_url PK
        TEXT state
        INTEGER last_check_at "nullable"
        INTEGER consecutive_failures
    }
    provider_configs {
        TEXT provider_id PK
        INTEGER timeout_ms
        TEXT preferred_mirror_url "nullable"
        INTEGER is_enabled
        INTEGER search_enabled
        INTEGER browse_enabled
        INTEGER download_enabled
        INTEGER use_anonymous
        TEXT sort_preference "nullable"
        INTEGER updated_at
    }
    provider_credential_binding {
        TEXT providerId PK
        TEXT credentialId
    }
    provider_credentials {
        TEXT provider_id PK
        TEXT auth_type
        TEXT username "nullable"
        TEXT encrypted_password "nullable"
        TEXT encrypted_token "nullable"
        TEXT encrypted_api_key "nullable"
        TEXT encrypted_api_secret "nullable"
        TEXT cookie_value "nullable"
        INTEGER expires_at "nullable"
        INTEGER is_active
        INTEGER last_used_at "nullable"
        INTEGER created_at
        INTEGER updated_at
    }
    provider_sync_toggle {
        TEXT providerId PK
        INTEGER enabled
    }
    Search {
        INTEGER id PK
        INTEGER timestamp
        TEXT query "nullable"
        TEXT sort
    }

```

```

        TEXT order
        TEXT period
        TEXT author "nullable"
        TEXT categories "nullable"
    }
    search_provider_selections {
        INTEGER id PK "autoincrement"
        TEXT query_hash
        TEXT provider_id
        INTEGER is_selected
        INTEGER created_at
    }
    Suggest {
        INTEGER id PK
        INTEGER timestamp
        TEXT suggest
    }
    sync_outbox {
        INTEGER id PK "autoincrement"
        TEXT kind
        TEXT payload
        INTEGER createdAt
    }
    tracker_mirror_user {
        TEXT tracker_id PK
        TEXT url PK
        INTEGER priority
        TEXT protocol
        INTEGER added_at
    }
    HistoryTopic {
        TEXT id PK
        INTEGER timestamp
        TEXT title
        TEXT author "nullable"
        TEXT category "nullable"
        TEXT tags "nullable"
        TEXT status "nullable"
        INTEGER date "nullable"
        TEXT size "nullable"
        INTEGER seeds "nullable"
        INTEGER leeches "nullable"
        TEXT magnetLink "nullable"
    }
}

```

1.2 Tables (client) — column listing

All columns below are exactly as declared in 11.json.

Bookmark — PK id

Column	Affinity	Not null
id	TEXT	yes
timestamp	INTEGER	yes
category	TEXT	yes
topics	TEXT	yes
newTopics	TEXT	yes

cloned_provider — PK syntheticId

Column	Affinity	Not null
syntheticId	TEXT	yes
sourceTrackerId	TEXT	yes
displayName	TEXT	yes
primaryUrl	TEXT	yes
deletedAt	INTEGER	no

credentials_entry — PK id

Column	Affinity	Not null
id	TEXT	yes
displayName	TEXT	yes
type	TEXT	yes
ciphertext	BLOB	yes
createdAt	INTEGER	yes
updatedAt	INTEGER	yes
deletedAt	INTEGER	no

Endpoint — PK id

Column	Affinity	Not null
id	TEXT	yes
type	TEXT	yes
host	TEXT	yes

FavoriteSearch — PK id

Column	Affinity	Not null
id	INTEGER	yes

FavoriteTopic — PK id

Column	Affinity	Not null
id	TEXT	yes
timestamp	INTEGER	yes
title	TEXT	yes
author	TEXT	no
category	TEXT	no
tags	TEXT	no
status	TEXT	no
date	INTEGER	no
size	TEXT	no
seeds	INTEGER	no
leeches	INTEGER	no
magnetLink	TEXT	no
hasUpdate	INTEGER	yes

ForumCategoryEntity — PK id; FK parentId → ForumCategoryEntity.id (ON DELETE CASCADE); index on parentId

Column	Affinity	Not null
id	TEXT	yes
name	TEXT	yes
parentId	TEXT	no
orderIndex	INTEGER	yes

ForumMetadata — PK id

Column	Affinity	Not null
id	INTEGER	yes
lastUpdatedTimestamp	INTEGER	yes

forum_provider_selections — PK id (AUTOINCREMENT)

Column	Affinity	Not null
id	INTEGER	yes
category_id	TEXT	yes
provider_id	TEXT	yes
created_at	INTEGER	yes

tracker_mirror_health — composite PK (tracker_id, mirror_url)

Column	Affinity	Not null
tracker_id	TEXT	yes
mirror_url	TEXT	yes
state	TEXT	yes
last_check_at	INTEGER	no
consecutive_failures	INTEGER	yes

provider_configs — PK provider_id

Column	Affinity	Not null
provider_id	TEXT	yes
timeout_ms	INTEGER	yes
preferred_mirror_url	TEXT	no
is_enabled	INTEGER	yes
search_enabled	INTEGER	yes
browse_enabled	INTEGER	yes
download_enabled	INTEGER	yes
use_anonymous	INTEGER	yes
sort_preference	TEXT	no
updated_at	INTEGER	yes

provider_credential_binding — PK providerId

Column	Affinity	Not null
providerId	TEXT	yes
credentialId	TEXT	yes

provider_credentials — PK provider_id

Column	Affinity	Not null
provider_id	TEXT	yes
auth_type	TEXT	yes
username	TEXT	no
encrypted_password	TEXT	no
encrypted_token	TEXT	no
encrypted_api_key	TEXT	no
encrypted_api_secret	TEXT	no
cookie_value	TEXT	no

Column	Affinity	Not null
expires_at	INTEGER	no
is_active	INTEGER	yes
last_used_at	INTEGER	no
created_at	INTEGER	yes
updated_at	INTEGER	yes

provider_sync_toggle — PK providerId

Column	Affinity	Not null
providerId	TEXT	yes
enabled	INTEGER	yes

Search — PK id

Column	Affinity	Not null
id	INTEGER	yes
timestamp	INTEGER	yes
query	TEXT	no
sort	TEXT	yes
order	TEXT	yes
period	TEXT	yes
author	TEXT	no
categories	TEXT	no

search_provider_selections — PK id (AUTOINCREMENT)

Column	Affinity	Not null
id	INTEGER	yes
query_hash	TEXT	yes
provider_id	TEXT	yes
is_selected	INTEGER	yes
created_at	INTEGER	yes

Suggest — PK id

Column	Affinity	Not null
id	INTEGER	yes
timestamp	INTEGER	yes

Column	Affinity	Not null
suggest	TEXT	yes

sync_outbox — PK id (AUTOINCREMENT)

Column	Affinity	Not null
id	INTEGER	yes
kind	TEXT	yes
payload	TEXT	yes
createdAt	INTEGER	yes

tracker_mirror_user — composite PK (tracker_id, url)

Column	Affinity	Not null
tracker_id	TEXT	yes
url	TEXT	yes
priority	INTEGER	yes
protocol	TEXT	yes
added_at	INTEGER	yes

HistoryTopic — PK id

Column	Affinity	Not null
id	TEXT	yes
timestamp	INTEGER	yes
title	TEXT	yes
author	TEXT	no
category	TEXT	no
tags	TEXT	no
status	TEXT	no
date	INTEGER	no
size	TEXT	no
seeds	INTEGER	no
leeches	INTEGER	no
magnetLink	TEXT	no

Room schema JSONs for all migration versions are checked in under `core/database/schemas/lava.database.AppDatabase/`. The build verifies that migrations between consecutive versions reproduce the next version's schema.

2. Server schema (Postgres, schema lava_api)

Tables live in the lava_api schema (created by migration 0001). Source files: [lava-api-go/migrations/](#). The one-shot lava-migrate compose service applies them with golang-migrate (see [docs/deployment/README.md](#)).

Migration history:

Migration	Table / change
0001	response_cache (initial) + lava_api schema
0002	request_audit
0003	rate_limit_bucket
0004	login_attempt
0005	response_cache realign (DROP + re-CREATE; destructive on existing DBs)
0006	provider_credentials
0007	provider_configs
0008	search_provider_selections
0009	forum_provider_selections

2.1 ER diagram (server)

No foreign keys are declared between these tables; each is standalone. The diagram lists the post-migration column shape (response_cache as realigned by 0005).

```
erDiagram
    response_cache {
        TEXT cache_key PK
        BYTEA value
        TIMESTAMPTZ expires_at "nullable"
    }
    request_audit {
        BIGSERIAL id PK
        TIMESTAMPTZ received_at
        TEXT method
        TEXT path
        TEXT query "nullable"
        INET client_ip "nullable"
        TEXT auth_realm_hash "nullable"
        SMALLINT upstream_status "nullable"
        INTEGER upstream_ms "nullable"
        TEXT cache_outcome
        INTEGER bytes_out "nullable"
    }
```

```

rate_limit_bucket {
    INET client_ip PK
    TEXT route_class PK
    DOUBLE tokens
    TIMESTAMPTZ last_refill_at
}
login_attempt {
    BIGSERIAL id PK
    INET client_ip
    TEXT username_hash
    BOOLEAN succeeded
    TIMESTAMPTZ attempted_at
}
provider_credentials {
    TEXT provider_id PK
    TEXT auth_type
    TEXT username "nullable"
    TEXT encrypted_password "nullable"
    TEXT encrypted_token "nullable"
    TEXT encrypted_api_key "nullable"
    TEXT encrypted_api_secret "nullable"
    TEXT cookie_value "nullable"
    TIMESTAMPTZ expires_at "nullable"
    BOOLEAN is_active
    TIMESTAMPTZ last_used_at "nullable"
    TIMESTAMPTZ created_at
    TIMESTAMPTZ updated_at
}
provider_configs {
    TEXT provider_id PK
    INTEGER timeout_ms
    TEXT preferred_mirror_url "nullable"
    BOOLEAN is_enabled
    BOOLEAN search_enabled
    BOOLEAN browse_enabled
    BOOLEAN download_enabled
    TEXT sort_preference "nullable"
    TIMESTAMPTZ updated_at
}
search_provider_selections {
    BIGSERIAL id PK
    TEXT query_hash UK
    TEXT provider_id UK
    BOOLEAN is_selected
    TIMESTAMPTZ created_at
}
forum_provider_selections {
    BIGSERIAL id PK
    TEXT category_id UK
    TEXT provider_id UK
    TIMESTAMPTZ created_at
}

```

2.2 Tables (server) — column listing

lava_api.response_cache — PK cache_key (post-0005 shape)

Column	Type	Constraint
cache_key	TEXT	PRIMARY KEY
value	BYTEA	NOT NULL
expires_at	TIMESTAMPTZ	—

Index: response_cache_expires_at_idx on expires_at. The submodules/cache/pkg/postgres library has these three column names hardcoded in its INSERT/SELECT (see the comment block in 0001/0005).

lava_api.request_audit — PK id

Column	Type	Constraint
id	BIGSERIAL	PRIMARY KEY
received_at	TIMESTAMPTZ	NOT NULL DEFAULT now()
method	TEXT	NOT NULL
path	TEXT	NOT NULL
query	TEXT	—
client_ip	INET	—
auth_realm_hash	TEXT	—
upstream_status	SMALLINT	—
upstream_ms	INTEGER	—
cache_outcome	TEXT	NOT NULL
bytes_out	INTEGER	—

Index: request_audit_received_at_idx on received_at.

lava_api.rate_limit_bucket — composite PK (client_ip, route_class)

Column	Type	Constraint
client_ip	INET	NOT NULL, PK
route_class	TEXT	NOT NULL, PK
tokens	DOUBLE PRECISION	NOT NULL
last_refill_at	TIMESTAMPTZ	NOT NULL

lava_api.login_attempt — PK id

Column	Type	Constraint
id	BIGSERIAL	PRIMARY KEY
client_ip	INET	NOT NULL
username_hash	TEXT	NOT NULL
succeeded	BOOLEAN	NOT NULL
attempted_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Index: login_attempt_lookup_idx on (client_ip, attempted_at DESC).

lava_api.provider_credentials — PK provider_id

Column	Type	Constraint
provider_id	TEXT	PRIMARY KEY
auth_type	TEXT	NOT NULL DEFAULT 'none'
username	TEXT	—
encrypted_password	TEXT	—
encrypted_token	TEXT	—
encrypted_api_key	TEXT	—
encrypted_api_secret	TEXT	—
cookie_value	TEXT	—
expires_at	TIMESTAMPTZ	—
is_active	BOOLEAN	NOT NULL DEFAULT true
last_used_at	TIMESTAMPTZ	—
created_at	TIMESTAMPTZ	NOT NULL DEFAULT now()
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Index: provider_credentials_active_idx on (is_active, updated_at DESC).

lava_api.provider_configs — PK provider_id

Column	Type	Constraint
provider_id	TEXT	PRIMARY KEY
timeout_ms	INTEGER	NOT NULL DEFAULT 10000
preferred_mirror_url	TEXT	—

Column	Type	Constraint
is_enabled	BOOLEAN	NOT NULL DEFAULT true
search_enabled	BOOLEAN	NOT NULL DEFAULT true
browse_enabled	BOOLEAN	NOT NULL DEFAULT true
download_enabled	BOOLEAN	NOT NULL DEFAULT true
sort_preference	TEXT	—
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

lava_api.search_provider_selections — PK id

Column	Type	Constraint
id	BIGSERIAL	PRIMARY KEY
query_hash	TEXT	NOT NULL
provider_id	TEXT	NOT NULL
is_selected	BOOLEAN	NOT NULL DEFAULT true
created_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Constraints: UNIQUE (query_hash, provider_id); index search_provider_selections_query_idx on query_hash.

lava_api.forum_provider_selections — PK id

Column	Type	Constraint
id	BIGSERIAL	PRIMARY KEY
category_id	TEXT	NOT NULL
provider_id	TEXT	NOT NULL
created_at	TIMESTAMPTZ	NOT NULL DEFAULT now()

Constraints: UNIQUE (category_id, provider_id); index forum_provider_selections_category_idx on category_id.

3. Room ↔ Postgres parity

The two databases serve different roles: the Room DB is the on-device client cache/state; the Postgres DB is the server-side cache/audit/credential store. Most tables exist on only one side. The conceptually overlapping entities:

Entity	Client (Room)	Server (Postgres)	Notes
provider_configs	yes (PK provider_id)	yes (PK provider_id)	Same intent (permissions, toggles + timeouts, preferred mirror). Column drift: the table has use_and_preferred_mirror (INTEGER, NOT NULL) which the Postgres table does not have (none since 2007 predates it). Both have is_enabled/search_enabled/browse_enabled/download_enabled/sort_preference/updated_at. Types differ by platform (Room uses INTEGER booleans, Postgres BOOLEANs, INTEGER epoch timestamps, TIMESTAMPTZ).
provider_credentials	yes (PK provider_id)	yes (PK provider_id)	Same column set and intent (auth_type, username, encrypted_cookie_value, expires, is_active, last_used). created_at, updated_at. Type encoding differs by platform (Room uses INTEGER vs Postgres TEXT/TIMESTAMP/BOOLEAN).
search_provider_selections	yes (PK id AUTOINCREMENT)	yes (PK id BIGSERIAL)	Same intent (permissions, provider selection). Both carry query_hash, provider_id, is_selected, created_at. The Postgres side adds a UNIQUE index (query_hash, provider_id) while the Room JSON column has no equivalent unique index.

Entity	Client (Room)	Server (Postgres)	Notes
forum_provider_selections	yes (PK id AUTOINCREMENT)	yes (PK id BIGSERIAL)	Same intent (per category provide selection). Room category_id, prov created_at. Postg UNIQUE (category_provider_id). Not Room table has is_selected column whereas search_provider_s (both sides) does

UNCONFIRMED: whether the client syncs any of these overlapping tables to the server (e.g. via the Room sync_outbox table) is not asserted here — it was not verified against sync code in this docs pass. The tables overlap in *shape and intent*; the synchronization mechanism (if any) is out of scope for this schema reference.

Client-only tables: Bookmark, cloned_provider, credentials_entry, Endpoint, FavoriteSearch, FavoriteTopic, ForumCategoryEntity, ForumMetadata, tracker_mirror_health, provider_credential_binding, provider_sync_toggle, Search, Suggest, sync_outbox, tracker_mirror_user, HistoryTopic.

Server-only tables: response_cache, request_audit, rate_limit_bucket, login_attempt.

4. Cross-references

- Room schema dump: [core/database/schemas/lava.database.AppDatabase/11.json](#)
- Postgres migrations: [lava-api-go/migrations/](#)
- API reference: [docs/api/README.md](#)
- Deployment guide: [docs/deployment/README.md](#)